Pre-assignment problem for unique minimum vertex cover on bounded clique-width graphs

Hongeun Im(GIST)

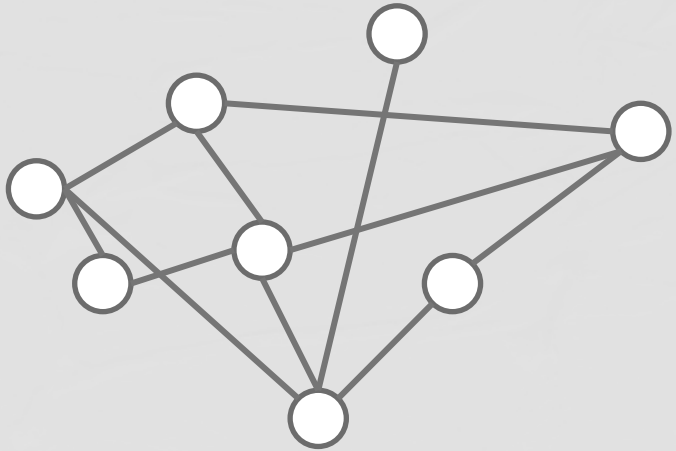
01

Motivation & Background

Problem Definition

Vertex Cover:

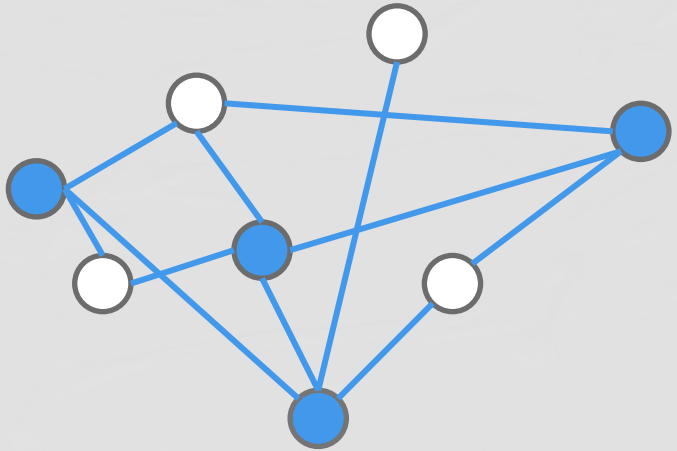
- A set S of vertices in a graph G is called a vertex cover of G if S meets all edges of G.



Problem Definition

Vertex Cover:

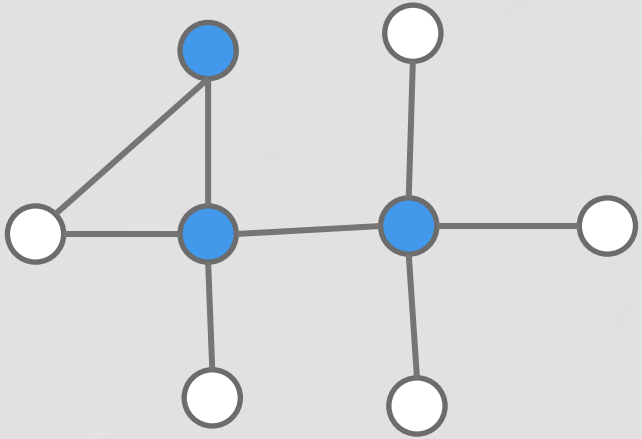
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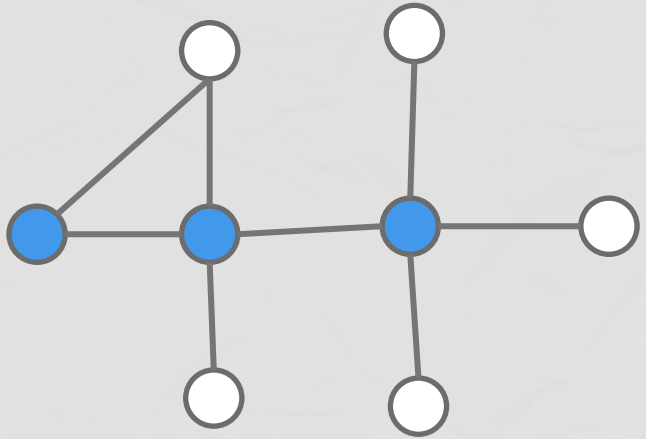


Problem Definition

Vertex Cover:

- A set S of vertices in a graph G is called a vertex cover of G if S meets all edges of G.
- There can be multiple minimum vertex covers





Problem Definition

Pre-assignment for uniquification of minimum vertex cover(PAU-VC)

- Input: A graph G and an integer k
- Is there a set S of at most k vertices in G such that among all minimum vertex covers of G, exactly one contains S?

Minimum pre-assignment for uniquification of minimum vertex cover(Min PAU-VC)

- Input: A graph G
- Find a minimum set S of vertices in G such that among all minimum vertex covers of G, exactly one contains S.

Motivation

Why do we need to solve the Min PAU-VC problem?

- Vertex cover problem is NP-complete
- Problems are easier if a unique solution is guaranteed
- We can guarantee a unique solution by finding a pre-assignment

Terminologies

Clique width

The minimum number of labels needed to construct G using four operations

- Create a vertex with label i
- Combine two labeled graphs(disjoint union)
- Add all possible edges between vertices labeled i and j ($i \neq j$)
- Relabel all vertices with label i to label j

Complete graphs have clique width≤ 2

Paths have clique width≤ 3

What will be the clique width of trees?

Ans:≤3

Terminologies

Fixed parameter tractable(FPT)

- Running time is $f(k) \cdot n^c$
- c is a constant, and the explosion is confined to $f(k)$.

Thm) Min PAU-VC is fixed-parameter tractable parameterized by clique-width.

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Polynomial algorithm for trees



Algorithm for trees

Notations

- $\mu_v(t)$=the size of the smallest vertex cover of G_v of type t(0 if v is included, 1 if v is not)
- β-set: a set $S \subseteq V(G_v)$ is a β-set if, for $\beta: \{0,1\} \to \{0,1,2\}$

if $\beta(0) \in \{0,1\}$, there are exactly $\beta(0)$ minimum vertex covers of G_v not containing v and containing S

if $\beta(0) = 2$, there are at least 2 minimum vertex covers of G_v not containing v and containing S

if $\beta(1) \in \{0,1\}$, there are exactly $\beta(1)$ minimum vertex covers of G_v containing v and containing S

if $\beta(1) = 2$, there are at least 2 minimum vertex covers of G_v containing v and containing S

Algorithm for trees

- We want: for every vertex v, for the 9 functions $\beta: \{0,1\} \to \{0,1,2\}$,
- Step 1: Compute $\mu_v(t)$

  t=0: $\mu_v(0) = \sum_w \mu_w(1)$, w are the children of v

  t=1: $\mu_v(0) = 1 + \sum_w \min(\mu_w(0), \mu_w(1))$

Algorithm for trees

- Step 2: Keep cheapest valid choice

  Try $v \in S$ and $v \notin S$

  For each child w, consider each possible β_w

  $\beta_v(0) = 0$ if $v \in S$, else $\min\left(2, \prod_w \beta_w(1)\right)$

  $$\beta_v(1) = \min\left(2, \prod_w \sum_{\text{t achieving } \min_\mu} \beta_w(t)\right)$$

  If there are multiple S that give the same β_v, keep cheapest(min($|S|$))

Algorithm for trees

- Step 3: Final S decision

  After calculating up to root r,

  $\mu = min(\mu_r(0), \mu_r(1))$

  Γ =types t with $\mu_r(t) = \mu$

  Get all S such that $\sum_{t\in\Gamma} \beta\,(t) = 1$.

  note: if $\mu_r(0) < \mu_r(1)$, then $\Gamma = \{0\}$, valid condition is $\beta(0) = 1$

  if $\mu_r(1) < \mu_r(0)$, then $\Gamma = \{1\}$, valid condition is $\beta(1) = 1$

  if $\mu_r(0) = \mu_r(1)$, then $\Gamma = \{0,1\}$, valid condition is $\beta(0) + \beta(1) = 1$

  If there are multiple S that satisfy, return smallest S

Algorithm for trees

- Time complexity

$O(v \times 2 \times |children(v)| \times 9 \times 9)$

$= O(E(G) \times 162)$

$= O\big((n-1) \times 162\big)$

$= O(n)$

Algorithm for tree is polynomial!

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Generalization to clique-width



Generalization to clique-width

Notations

- full_H(x): set of labels $i \in [k]$ where $\text{lab}_H^{-1}(i) \subseteq X$

  =labels such that all corresponding vertices are in X

- min vertex cover with respect to $I \subseteq [k]$: minimum among vertex covers X with $\text{full}_H(X) = I$

  =min vertex cover that has all vertices with label $i \in I$ included

Generalization to clique-width

- Lemma 3.1. Let k be a positive integer. Given a k-labeled graph (G, lab_G) with $\prod(G)$, μ_G and a collection of minimum β-sets for $\beta \in \prod(G)$, one can solve MIN PAU-VC for G in time $2^{O(2^k)}|V(G)|$.
- pf: Let $\mu = \min_{I \subseteq [k]} \mu_G(I)$, and $\Gamma = \{J \subseteq [k] : \mu_G(J) = \mu\}$. Then μ is the size of a minimum vertex cover of G. We say that a function $\beta : 2^{[k]} \to \{0,1,2\}$ is valid if $\sum_{J \in \Gamma} \beta(J) = 1$. A β-set with a valid function $\beta \in \Pi(G)$ is a set forcing a unique minimum vertex cover in G. Thus, the minimum β-set with a valid function $\beta \in \Pi(G)$ is a required solution for Min PAU-VC.
- note: We want to find S with characteristic β such that $\sum_{J \in \Gamma} \beta(J) = 1$. This means that there is one minimum vertex cover(with respect to any J) containing S.

Generalization to clique-width

Intuition

- Tree algorithm
- $\beta: \{0,1\} \rightarrow \{0,1,2\}$
- $\{0,1\}$ means we check v included/not included in minimum vertex covers.
- Only v has neighbors outside T_v. V being included/not included affects whether the children of v has to be in the vertex cover.
- T_v: subtree rooted at v

- Clique-width
- $\beta: 2^{[k]} \rightarrow 0,1,2$
- $2^{[k]}$ is the powerset of $[k]$(i.e. the set of all subsets of $[k]$). We check whether labels $i \in I$ are completely included for all $I \subseteq [k]$.
- Vertices of label i being completely in a vertex cover or not affects whether the incoming vertices of label j has to be in the vertex cover.
- H: k-labeled graph at some point of the construction with clique-width construction rules.

Q&A

References

- An, S., Chang, Y., Cho, K., Kwon, O., Lee, M., Oh, E., & Shin, H. (2026). *Pre-assignment problem for unique minimum vertex cover on bounded clique-width graphs.* Theoretical Computer Science, Volume 1084, Article 116173, ISSN 0304-3975, https://doi.org/10.1016/j.tcs.2026.116173